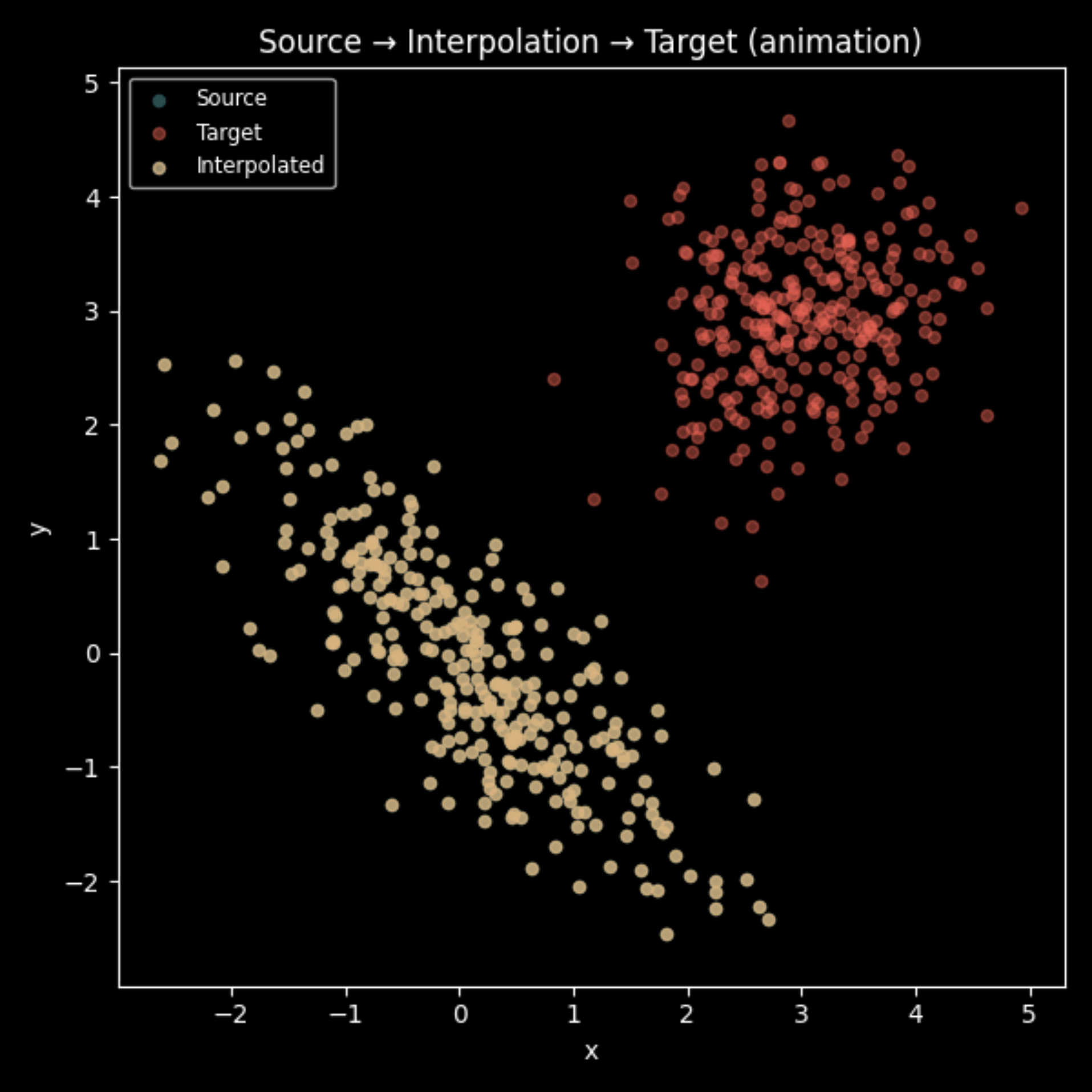
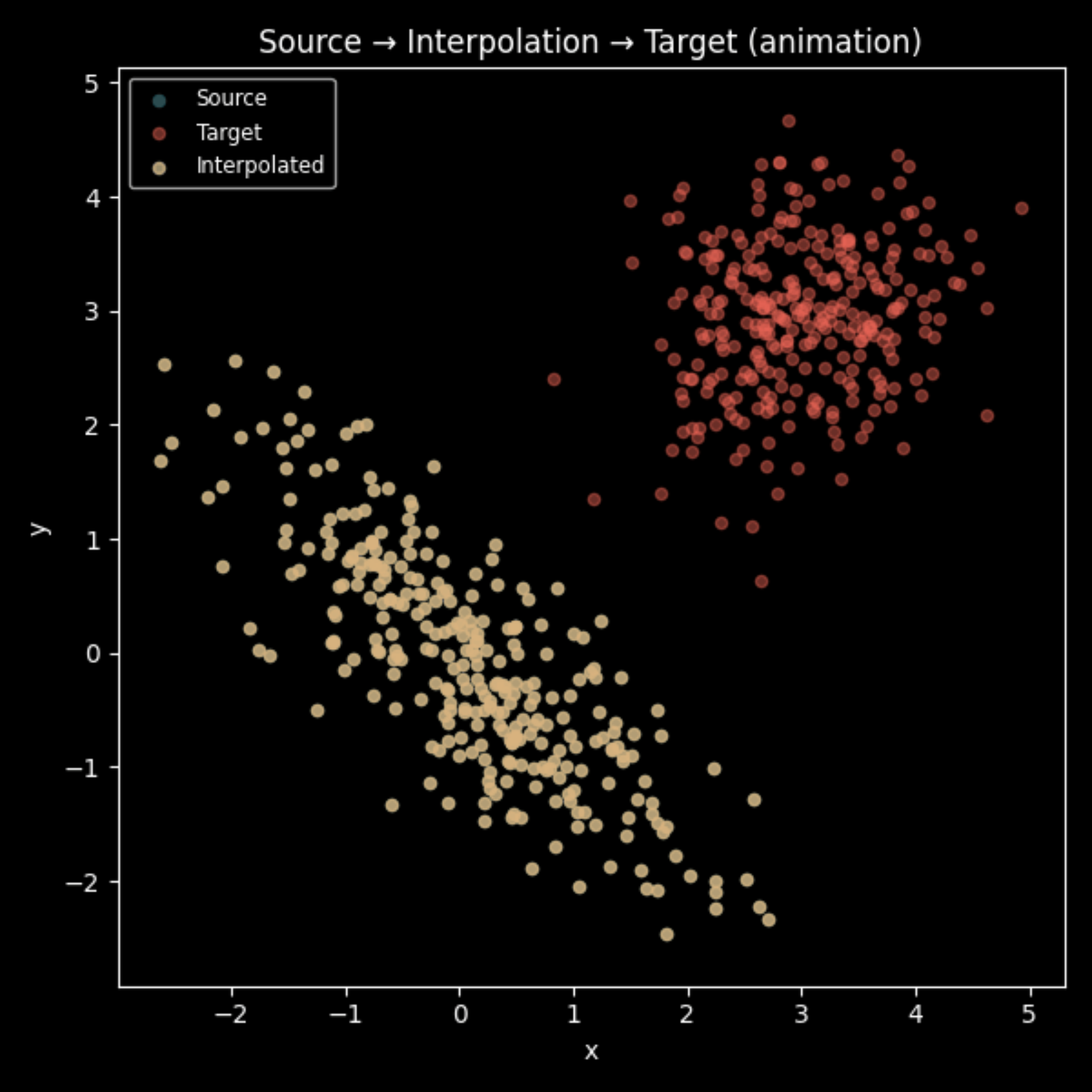

Source → Interpolation → Target (animation)
Source
Target
Interpolated
y
x
5
4
3
2
1
0
−1
−2
−2
−1
0
1
2
3
4
5

```python
from scipy.linalg import sqrtm
from numpy.linalg import inv

def gaussian_ot_map(X, μ1, μ2, Σ1, Σ2):
    """Returns Monge map T(X) from source to target
       Gaussian under squared cost."""
    sqrt_Σ1 = sqrtm(Σ1)
    inv_sqrt_Σ1 = inv(sqrt_Σ1)
    A = inv_sqrt_Σ1 @ sqrtm(sqrt_Σ1 @ Σ2 @ sqrt_Σ1) @ inv_sqrt_Σ1
    return μ2 + (X - μ1) @ A.T

# Sample Gaussians
μ1, Σ1 = [0, 0], [[1, -0.8], [-0.8, 1]]
μ2, Σ2 = [3, 3], [[0.5, 0.1], [0.1, 0.5]]
X = np.random.multivariate_normal(μ1, Σ1, 300)
Y = np.random.multivariate_normal(μ2, Σ2, 300)
TX = gaussian_ot_map(X, μ1, μ2, Σ1, Σ2)

# Interpolation
alphas = np.linspace(0, 1, 21)
interpolated = [(1 - a) * X + a * TX for a in alphas]
```


Source → Interpolation → Target (animation)
Source
Target
Interpolated
y
x
5
4
3
2
1
0
−1
−2
−2
−1
0
1
2
3
4
5

Interactive Demo: Gaussian Monge

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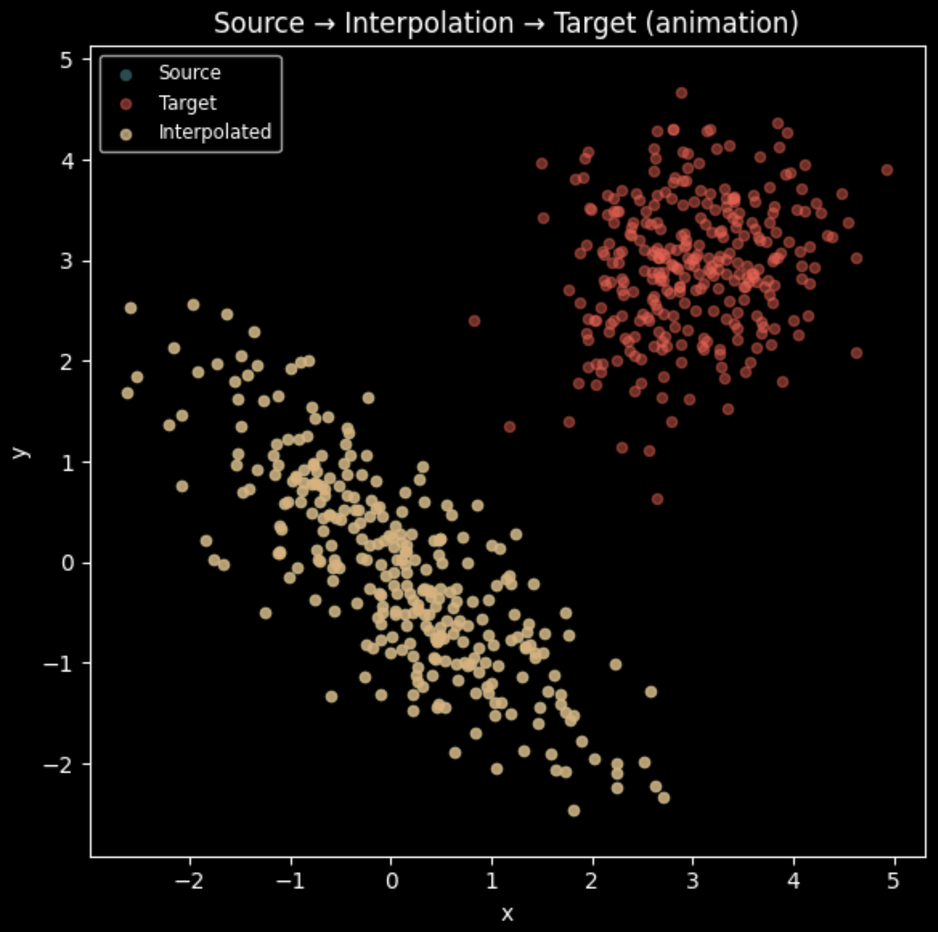

Next Time

- Relaxing the problem, Kanotorvich formulation, couplings
- Reading: Peyré & Cuturi (2019), Sections 2 & 3
- Scribing signup released tomorrow